

19	A	Option A: Correct $\theta = \frac{\lambda}{b} \Rightarrow b = \frac{\lambda}{\theta} = \frac{590 \times 10^{-9}}{2.0 \times 10^{-6}} = 0.295 \text{ m} \approx 0.30 \text{ m}$ Option B: Incorrect Red stars give out red light of wavelength of 700 nm. $\theta = \frac{\lambda}{b} = \frac{700 \times 10^{-9}}{0.30} = 2.3 \times 10^{-6} \text{ radians}$ which is larger than the angular separation of 2.0×10^{-6} radians. Hence, the red stars cannot be resolved. Option C: Incorrect The resolving power of the telescope is dependent of the wavelength of light, according to the expression $\theta = \frac{\lambda}{b}$. Option D: Incorrect The angular separation between the stars is 2.0×10^{-6} radians for the telescope to be able to resolve. Hence, the telescope cannot be used to distinguish the stars.
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Qn	Ans	Solution
20	A	For the first 2.0 m, the movement is along the direction of increasing potential